Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

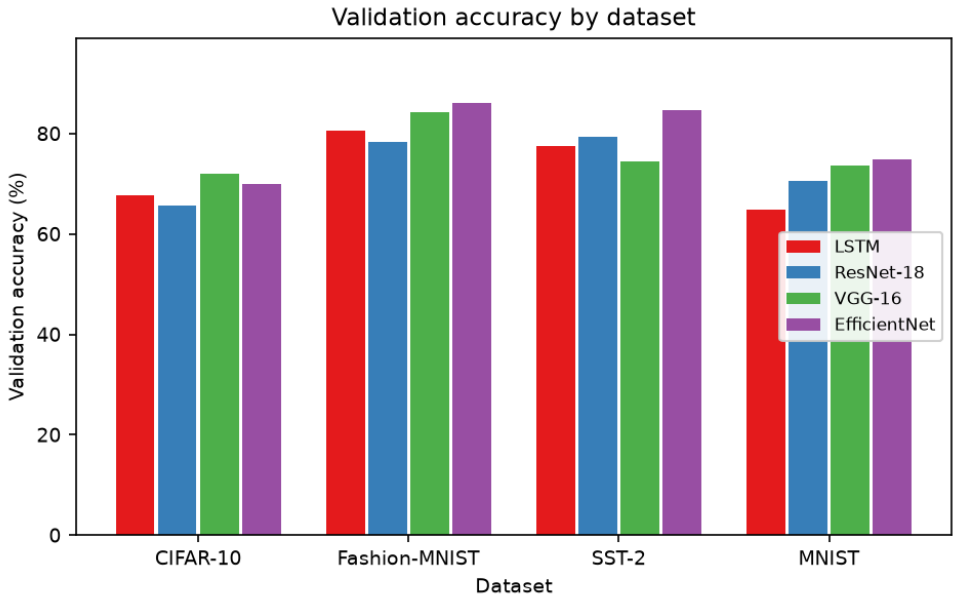


Figure 1: Validation accuracy by dataset for the models in this experiment.

On SST-2, ResNet-18 is 51.3 points better than LSTM. VGG-16 reaches about 70% validation accuracy on SST-2.

Figure 2 shows the training loss.

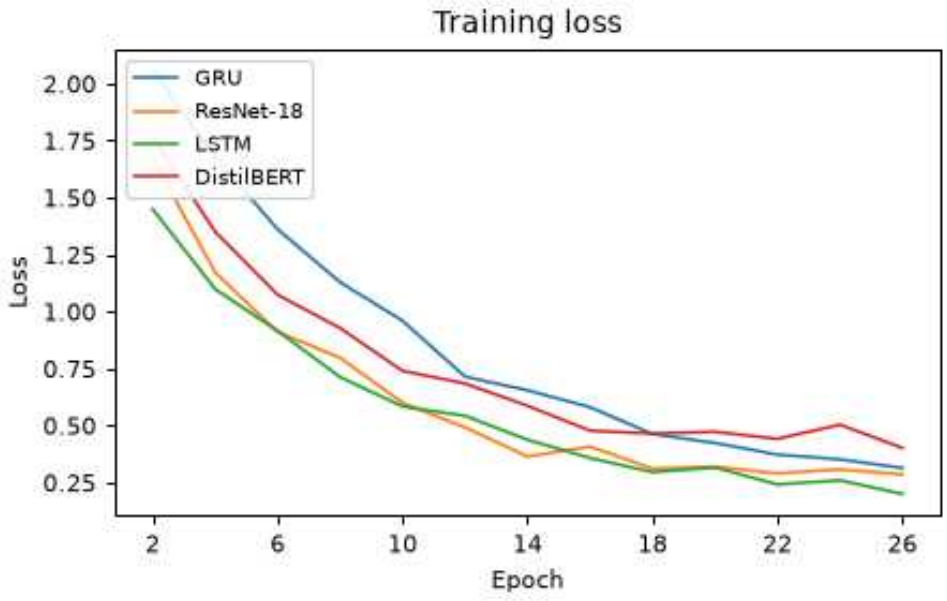


Figure 2: Training loss for the models in this experiment.

LSTM has a loss of 0.464 at epoch 24. At epoch 18, the loss of GRU is 0.46. GRU has a loss of 0.257 at epoch 18 (Figure 2).

Figure 3 shows the f1 score by dataset.

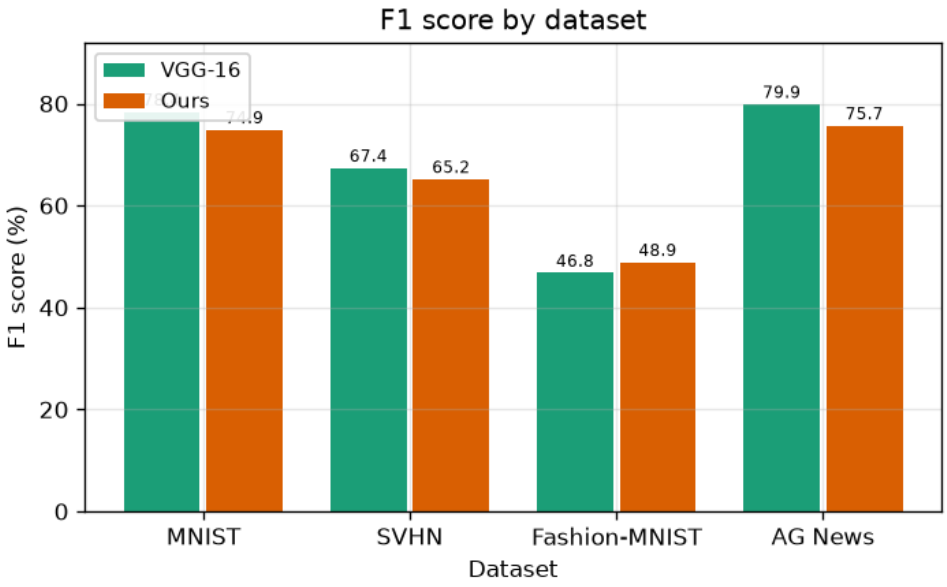


Figure 3: F1 score by dataset for the models in this experiment.

On AG News, VGG-16 is about 4 points better than Ours (Figure 3). Ours achieves over 60% F1 score on SVHN. VGG-16 reaches 75.3% F1 score on AG News.

3 Discussion

XGBoost reaches 93.2% accuracy on SVHN. The differences between the models are discussed in the next section.